Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-II-Q05] | Part A (Q1-Q5)

Triangle ABC has angles $A = 84^\circ, B = 60^\circ, C = 36^\circ$, and side midpoints $D \in BC, E \in CA, F \in AB$. Their circle meets BD, AE, AF again at G, H, J. The six cyclic points occur as G, D, E, H, J, F. Find the sum of consecutive minor-arc measures $\widehat{DE} + 2\widehat{HJ} + 3\widehat{FG}$.

三角形 ABC 的角为 $84^\circ, 60^\circ, 36^\circ$，$D, E, F$ 分别为 BC, CA, AB 中点。其外接圆与 BD, AE, AF 再交于 G, H, J，六点沿圆依次为 G, D, E, H, J, F。求相邻劣弧度数 $\widehat{DE} + 2\widehat{HJ} + 3\widehat{FG}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-II-Q06] | Part B (Q6-Q10)

A square $ABCD$ is inscribed in a circle. Another square $EFGH$ has E, F on segment CD and G, H on the circle. Its area ratio to $ABCD$ is $m/n < 1$ reduced. Find $10n + m$.

正方形 $ABCD$ 内接于圆。另一正方形 $EFGH$ 的 E, F 位于线段 CD，G, H 位于圆上。其面积与原正方形之比为最简分数 $m/n < 1$，求 $10n + m$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2011-II-Q10] | Part B (Q6-Q10)

A radius-25 circle has intersecting chords of lengths $30, 14$ with midpoints 12 apart. Their intersection is P, and the center is O. Express $OP^2 = m/n$ in lowest terms and find $(m + n) \bmod 1000$.

半径二十五的圆中，两条长为三十、十四的弦交于 P，两弦中点相距十二。圆心为 O。若 $OP^2 = m/n$ 已最简，求 $m + n$ 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2013-II-Q08] | Part B (Q6-Q10)

A cyclic hexagon has successive sides $22, 22, 20, 22, 22, 20$. If its circumradius is $p + \sqrt{q}$ for positive integers, find $p + q$.

一个圆内接六边形的边依次为 $22, 22, 20, 22, 22, 20$。若外接圆半径为 $p + \sqrt{q}$，求正整数 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2015-I-Q06] | Part B (Q6-Q10)

On one circle, A, B, C, D, E occur consecutively at equal arc intervals along a minor arc. On a second circle centered at C, E, F, G, H, I, A occur consecutively at equal arc intervals along a minor arc. Given $\angle ABD - \angle AHG = 12^\circ$, determine $\angle BAG$ in degrees.

第一圆的小弧上，A, B, C, D, E 依次等弧分布。第二圆以 C 为圆心，其小弧上 E, F, G, H, I, A 依次等弧分布。已知 $\angle ABD - \angle AHG = 12^\circ$，求 $\angle BAG$ 的度数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-II-Q10] | Part B (Q6-Q10)

Triangle ABC is cyclic. Along side AB, the points occur as A, P, Q, B with $AP = 4, PQ = 3, QB = 6$. Rays CP, CQ reach the circumcircle again at S, T. Given $AS = 7, BT = 5$, express $ST = m/n$ in lowest terms and find $m + n$.

三角形 ABC 内接于圆。边 AB 上点序为 A, P, Q, B，$AP = 4, PQ = 3, QB = 6$。射线 CP, CQ 再次交外接圆于 S, T。已知 $AS = 7, BT = 5$，将 $ST = m/n$ 约为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2023-II-Q09] | Part B (Q6-Q10)

Two intersecting circles meet at P, Q. Their common external tangent nearer P touches them at A, B. A line through P parallel to AB meets the circles again at X, Y, with $PX = 10, PY = 14, PQ = 5$. The area of trapezoid $XABY$ is $m\sqrt{n}$ with squarefree n. Find $m + n$.

两圆交于 P, Q，离 P 较近的公外切线分别切于 A, B。过 P 作 AB 的平行线，再交两圆于 X, Y，其中 $PX = 10, PY = 14, PQ = 5$。梯形 $XABY$ 面积为 $m\sqrt{n}$，n 无平方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2024-I-Q10] | Part B (Q6-Q10)

Triangle ABC has $AB = 5, BC = 9, CA = 10$. Tangents to its circumcircle at B, C meet at D, and segment AD meets the circle again at P. Express $AP = m/n$ in lowest terms and find $m + n$.

三角形 ABC 中 $AB = 5, BC = 9, CA = 10$。外接圆在 B, C 处的切线交于 D，线段 AD 与圆再交于 P。将 AP 化为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-I-Q06] | Part B (Q6-Q10)

An isosceles trapezoid of area 72 has an incircle of radius 3. Its unequal parallel sides have lengths r, s. Find $r^2 + s^2$.

等腰梯形面积为 72，内切圆半径为 3，两条不等的平行边长为 r, s。求 $r^2 + s^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2026-II-Q10] | Part B (Q6-Q10)

In triangle ABC with $AB = 200, AC = 225$, the internal angle bisector AD meets BC at D. A circle through A tangent to BC at D meets AB, AC again at interior points E, F. If AE, AF, BD, CD are positive integers, maximize BC.

三角形 ABC 中 $AB = 200, AC = 225$，内角平分线 AD 交 BC 于 D。过 A 且在 D 与 BC 相切的圆再与 AB, AC 的内部交于 E, F。若 AE, AF, BD, CD 均为正整数，求最大 BC。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-I-Q13] | Part C (Q11-Q15)

In one circle the chord for an arc of d degrees has length 22, and the chord for $2d$ degrees is 20 longer than that for $3d$ degrees, with $0 < d < 120$. Write the $3d$ chord length as $-m + \sqrt{n}$ for positive integers m, n. Find $m + n$.

同一圆中，d 度弧的弦长为 22，$2d$ 度弧的弦比 $3d$ 度弧的弦长 20，且 $0 < d < 120$。将 $3d$ 弦长写为 $-m + \sqrt{n}$，求正整数 m, n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2002-I-Q13] | Part C (Q11-Q15)

Triangle ABC has medians $AD = 18, CE = 27$ and side $AB = 24$. Extend median CE past E to meet the circumcircle again at F. If $[AFB] = m\sqrt{n}$ with squarefree n, find $m + n$.

三角形 ABC 中中线 $AD = 18, CE = 27$，且 $AB = 24$。将 CE 越过 E 延长，再交外接圆于 F。若 $[AFB] = m\sqrt{n}$，n 无平方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2005-I-Q15] | Part C (Q11-Q15)

Triangle ABC has $AB = 20$. Its incircle cuts the median from C into three equal segments. If the area is $m\sqrt{n}$ with squarefree n, find $m + n$.

三角形 ABC 中 $AB = 20$，内切圆将从 C 出发的中线分为三等段。若面积为 $m\sqrt{n}$ 且 n 无平方素因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2009-I-Q15] | Part C (Q11-Q15)

Triangle ABC has $AB = 10, BC = 14, CA = 16$. For interior $D \in BC$, let I_B, I_C be incenters of ABD, ACD. The circles through B, I_B, D and C, I_C, D meet again at P. The largest area $[BPC]$ is $a - b\sqrt{c}$ in squarefree form. Find $a + b + c$.

三角形 ABC 边长为十、十四、十六。D 在 BC 内，I_B, I_C 为 ABD, ACD 内心。圆 BI_BD, CI_CD 再交于 P。BPC 面积最大值为 $a - b\sqrt{c}$，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2010-II-Q15] | Part C (Q11-Q15)

Triangle ABC has $AC = 13, BC = 14, AB = 15$. Let M, N be the midpoints of AC, AB, and let the internal bisectors from B, C meet those sides at D, E. The circles through A, M, N and A, D, E meet again at P. Ray AP meets BC at Q. If $BQ/CQ = m/n$ in lowest terms, find $m - n$.

三角形 ABC 中 $AC = 13, BC = 14, AB = 15$，$M, N$ 为 AC, AB 中点，B, C 的内角平分线分别交这两边于 D, E。圆 AMN 与圆 ADE 再交于 P，射线 AP 交 BC 于 Q。将 BQ/CQ 约为 m/n，求 $m - n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2012-II-Q15] | Part C (Q11-Q15)

Triangle ABC with sides $AB = 5, BC = 7, AC = 3$ has circumcircle ω. The A-bisector meets BC at D and ω again at E. The circle of diameter DE meets ω again at F. Reduce $AF^2 = m/n$ and find $m + n$.

三角形 ABC 的边长为 $AB = 5, BC = 7, AC = 3$，外接圆为 ω。A 的角平分线交 BC 于 D、再交 ω 于 E。以 DE 为直径的圆再交 ω 于 F。将 AF^2 约为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2014-I-Q15] | Part C (Q11-Q15)

In triangle ABC, $AB = 3, BC = 4, AC = 5$. A circle passes through B, meets AB again at E, BC again at D, and AC at distinct F, G, with all named points on the side segments. Given $EF = DF, DG/EG = 3/4$, express $DE = a\sqrt{b}/c$ with coprime a, c and squarefree b, and find $a + b + c$.

三角形 ABC 的边为 $AB = 3, BC = 4, AC = 5$。过 B 的圆再交 AB, BC 于 E, D，交 AC 于不同的 F, G，各点均在对应边线段上。若 $EF = DF, DG/EG = 3/4$，将 DE 写为 $a\sqrt{b}/c$，a, c 互质、b 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-I-Q15] | Part C (Q11-Q15)

Two intersecting circles meet at X, Y and share an external tangent touching them at A, B. The tangent line is nearer X than Y. A circle through A, B meets the first circle again at D and the second again at C, with C, Y, D collinear. Given $XC = 67, XY = 47, XD = 37$, find AB^2.

两相交圆交于 X, Y，一条外公切线分别切于 A, B，且此切线距 X 比距 Y 近。过 A, B 的另一圆再次交第一圆于 D、第二圆于 C，并且 C, Y, D 共线。已知 $XC = 67, XY = 47, XD = 37$，求 AB^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-I-Q15] | Part C (Q11-Q15)

Four distinct stick lengths form three noncongruent convex cyclic quadrilaterals, all with circumradius one and common area K. Their acute diagonal angles have sines $2/3, 3/5, 6/7$. Reduce $K = m/n$ and find $m + n$.

四根不同长度的杆可构成三个不全等凸圆内接四边形，外接圆半径均为一，公共面积为 K。三个对角线锐夹角的正弦分别为 $2/3, 3/5, 6/7$。将 K 约为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-II-Q14] | Part C (Q11-Q15)

The incircle of ABC touches BC at X. Line AX meets it again at $Y \neq X$. The tangent at Y meets side segments AB, AC at P, Q. Given $AP = 3, PB = 4, AC = 8$, reduce $AQ = m/n$ and find $m + n$.

三角形内切圆切 BC 于 X，直线 AX 再交圆于 $Y \neq X$。Y 处切线交边线段 AB, AC 于 P, Q。已知 $AP = 3, PB = 4, AC = 8$，将 AQ 约为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q13] | Part C (Q11-Q15)

Triangle ABC has $AB = 4, BC = 5, CA = 6$. Points D, E on ray AB satisfy $AB < AD < AE$. The circles ACD, EBC meet again at $F \neq C$ with $DF = 2, EF = 7$. Write $BE = (a + b\sqrt{c})/d$ reduced with squarefree c, and find $a + b + c + d$.

三角形 ABC 边长为 $AB = 4, BC = 5, CA = 6$。射线 AB 上 $AB < AD < AE$，圆 ACD, EBC 再交于 $F \neq C$，且 $DF = 2, EF = 7$。将 $BE = (a + b\sqrt{c})/d$ 约简，求 $a + b + c + d$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q15] | Part C (Q11-Q15)

A chord AB of circle ω contains P, with $AP = 5, PB = 3$. Inner circles through A, P and through B, P are each internally tangent to ω, and meet again at Q. Their common chord line meets ω at X, Y, with $XY = 11$. Write $PQ^2 = m/n$ reduced and find $m + n$.

圆 ω 的弦 AB 上有 P，$AP = 5, PB = 3$。分别过 A, P 及 B, P 的两内圆与大圆内切，并再交于 Q。公共弦直线与大圆交于 X, Y，$XY = 11$。若 $PQ^2 = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-II-Q15] | Part C (Q11-Q15)

In acute triangle ABC, let P be the foot from C to AB and Q the foot from B to AC. Their line meets the circumcircle at X, Y, with $XP = 10, PQ = 25, QY = 15$. Write $AB \cdot AC = m\sqrt{n}$ with squarefree n and find $m + n$.

锐角三角形 ABC 中，P 为 C 到 AB 垂足，Q 为 B 到 AC 垂足。PQ 所在直线与外接圆交于 X, Y，$XP = 10, PQ = 25, QY = 15$。若 $AB \cdot AC = m\sqrt{n}$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2020-II-Q15] | Part C (Q11-Q15)

For an acute scalene triangle ABC, tangents to its circumcircle at B, C meet at T. Drop perpendiculars from T to the lines AB, AC, with feet X, Y. Given $TB = TC = 16$, $BC = 22$, and $TX^2 + TY^2 + XY^2 = 1143$, find XY^2.

锐角不等边三角形 ABC 的外接圆在 B, C 处的切线交于 T。从 T 向直线 AB, AC 作垂线，垂足为 X, Y。已知 $TB = TC = 16$、$BC = 22$、$TX^2 + TY^2 + XY^2 = 1143$，求 XY^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-I-Q11] | Part C (Q11-Q15)

A cyclic quadrilateral has successive side lengths $4, 5, 6, 7$. Project each vertex perpendicularly onto the other diagonal, obtaining $A_1, C_1 \in BD$ and $B_1, D_1 \in AC$. If the perimeter of $A_1B_1C_1D_1$ is m/n reduced, find $m+n$.

圆内接四边形 $ABCD$ 连续边长为 $4, 5, 6, 7$。将 A, C 向 BD 投影得 A_1, C_1，将 B, D 向 AC 投影得 B_1, D_1。若 $A_1B_1C_1D_1$ 周长约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-II-Q14] | Part C (Q11-Q15)

An acute triangle has circumcenter O and centroid G. The tangent at A meets the perpendicular to GO through G at X; let $Y = XG \cap BC$. Given $\angle B : \angle C : \angle XOY = 13 : 2 : 17$, write $\angle A = m/n$ degrees reduced and find $m + n$.

锐角三角形外心为 O、重心为 G。A 处切线与过 G 且垂直 GO 的直线交于 X，令 $Y = XG \cap BC$。已知 $\angle B : \angle C : \angle XOY = 13 : 2 : 17$。若 $\angle A$ 的度数约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-I-Q11] | Part C (Q11-Q15)

In parallelogram $ABCD$ with acute $\angle BAD$, a circle is tangent to side segments DA, AB, BC. Its intersections with diagonal AC occur in order A, P, Q, C, with $AP = 3, PQ = 9, QC = 16$. Write the area as $m\sqrt{n}$, where n is squarefree, and find $m + n$.

平行四边形 $ABCD$ 的 $\angle BAD$ 为锐角。一个圆与边线段 DA, AB, BC 相切，并与对角线 AC 依次交于 P, Q；A, P, Q, C 共线且依此排列，$AP = 3, PQ = 9, QC = 16$。面积写成 $m\sqrt{n}$，其中 n 无平方因子。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-II-Q15] | Part C (Q11-Q15)

Externally tangent circles have centers O_1, O_2 with $O_1O_2 = 15$. A third circle through both centers meets the first at B, C and the second at A, D. The cyclic order A, B, O_1, C, D, O_2 forms a convex hexagon. Given $AB = 2, CD = 16$, find its area.

两外切圆的圆心为 O_1, O_2，$O_1O_2 = 15$。第三圆经过两圆心，与第一圆交于 B, C，与第二圆交于 A, D。六点按 A, B, O_1, C, D, O_2 顺序构成凸六边形。已知 $AB = 2, CD = 16$，求面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程